

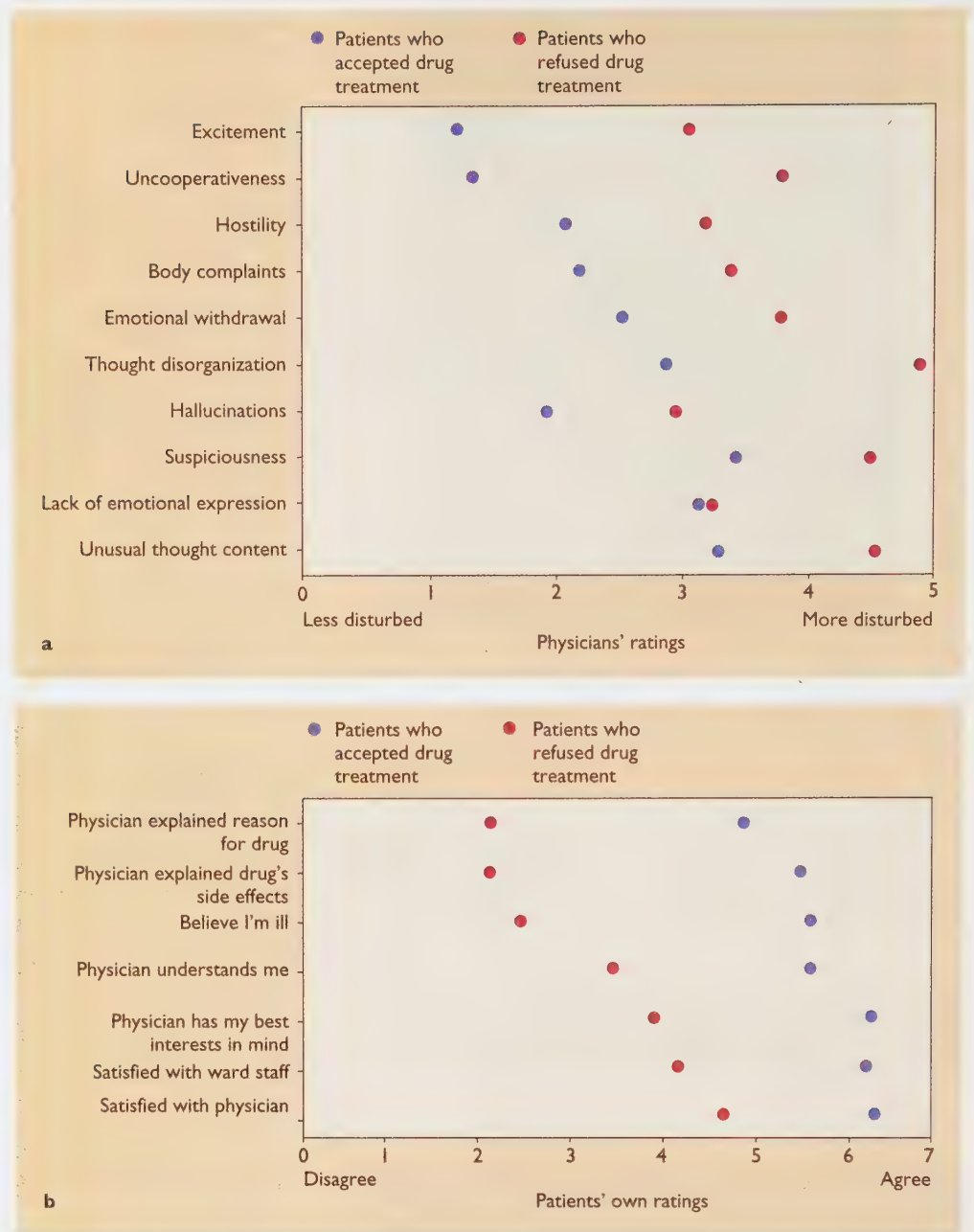


FIGURE 15.10 People with schizophrenia who refuse drug therapy impress their physicians as being seriously disturbed. The patients who refuse drugs rate themselves as dissatisfied with their physicians and their treatments. (a) Physicians' ratings of their patients. High scores indicate greater disturbance. Those refusing treatment showed greater indications of disturbance on most scales. (b) Patients' self-ratings. The higher scores of patients who agreed to drug treatment indicate their higher levels of satisfaction with their treatment. (Based on data from Marder et al., 1983.)

Before you answer “yes,” consider the following: First, psychotherapy is based on a trusting relationship in which the therapist promises to keep secret anything that the client says. Second, therapists are seldom certain about which clients are really dangerous (Monahan, 1993). Suppose you are a therapist. Over the course of a day, you see eight patients, and three of them say they are so angry they could kill somebody. How do you decide which ones, if any, to take seriously?

In 1976, a California court ruled in the **Tarasoff** case that a therapist who knew, or who should have known, that a client was dangerous is obligated to break the pledge of confidentiality and warn the endangered person (Weiner & Wettstein, 1993). In 1991, a Canadian court made a similar ruling in the case of *Wenden v. Trikha* (Truscott & Crook, 1993). However, applying this rule in practice is difficult, because of the uncertainty in judging dangerousness. The

courts have also ruled that the duty to inform applies only to an identifiable endangered individual. If the client is, for example, an alcohol- and drug-abusing airplane pilot, countless people are endangered, but the therapist is not obligated to inform anyone. In fact, the rules on confidentiality prohibit the therapist from informing the pilot's employer (Feltous, 1993). One can sympathize with therapists who wonder what to do, short of keeping a legal consultant available. Fortunately, the courts have so far been reluctant to hold therapists legally accountable in uncertain cases.

The Insanity Defense

Suppose, in the midst of an epileptic seizure, you flail your arms about and accidentally knock someone down the stairs. If that person dies, should you be convicted of murder? Of

course not. Now, suppose that, in the midst of severely disordered thinking and perception, you attack what appears to you to be a giant insect, but is in fact a human being. Should you be convicted of murder in that case?

Courts in North America and most European countries have traditionally ruled that you are “not guilty by reason of insanity.” Most people agree with that principle, at least for extreme cases. The problem is where to draw the line. Under what conditions should someone be judged legally insane? Insanity is a legal term, not a psychological or medical one, and its definition is based at least as much on politics as it is on science.

One point of broad agreement is that the crime itself, no matter how atrocious, does not demonstrate insanity. In Chapter 5, we considered the case of the “Hillside Strangler” who committed a long series of rapes and murders but who was eventually ruled sane and sentenced to prison. The same decision was made for Jeffrey Dahmer, who was arrested in 1991 after murdering and cannibalizing several men. Theodore Kaczynski, arrested as the “Unabomber” who had mailed bombs to many people for decades, refused to plead insanity, and probably would not have been ruled insane anyway. Bizarre crimes do not, in themselves, demonstrate insanity. In fact, each of these murderers knew what he was doing and took steps to avoid getting caught.

In many cases, however, a decision about sanity or insanity is difficult; lawyers, physicians, and psychologists have long struggled to establish a clear and acceptable definition of *insanity*. Probably the most famous definition, the **M’Naghten rule**, written in Great Britain in 1843, states:

To establish a defense on the ground of insanity, it must be clearly proved that, at the time of the committing of the act, the party accused was laboring under such a defect of reason, from disease of the mind, as not to know the nature and quality of the act he was doing; or if he did know it, that he did not know he was doing what was wrong. (Shapiro, 1985)

In other words, to be regarded as insane under the M’Naghten rule, people must be so disordered that they do not realize they are committing a criminal act. Presumably,

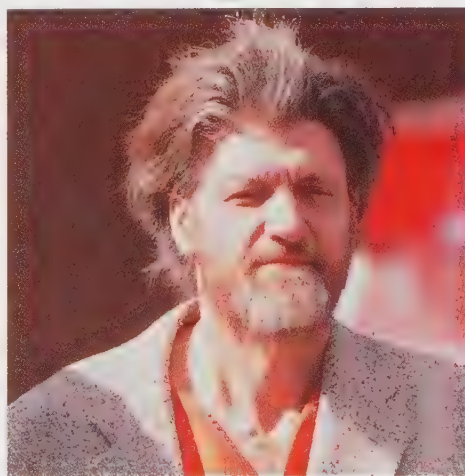
they would continue the act even if a police officer were standing nearby. Many observers have considered that rule too narrow and would like to broaden the definition of insanity to include “irresistible impulses”—acts similar to sneezing or hiccupping, which people could not inhibit for long, no matter how hard they tried. Under the **Durham rule**, established in 1954 in the case of a man named Durham, a U.S. court held that a defendant is not criminally responsible if the activity was “a product of mental disease or defect.” That rule confused more than it clarified and is no longer used. (Is ignorance a mental defect? What about sleepiness? grouchiness? Almost any criminal act is the product of some sort of defect, broadly defined.)

The **Model Penal Code**, written by the American Law Institute in the 1950s, offered another definition:

A person is not responsible for criminal conduct if at the time of such conduct as a result of mental disease or defect he lacks substantial capacity either to appreciate the criminality (wrongfulness) of his conduct or to conform his conduct to the requirements of law. (Simon, 1994)

This rule is not as loose as the Durham rule but not as rigid as the M’Naghten rule. However, a jury still needs to make a difficult judgment of whether a defendant lacked “substantial capacity” to appreciate wrongfulness or to conform his or her behavior to the law. When a jury finds someone insane in a highly publicized case—as in the case of John Hinckley, who attempted to assassinate President Ronald Reagan—many people are outraged, partly because they fear that the person will be quickly released. That fear is mostly misinformed; on the average, people found not guilty by reason of insanity are confined to a mental hospital for about as long as they would have spent in prison if they had been found guilty (Silver, 1995).

Several states in the United States have experimented with allowing a verdict of “guilty but mentally ill,” intended as a compromise between guilty and not guilty. However, the defendant actually receives the same sentence as if he or she had been simply found guilty and is no more likely to receive psychiatric care in prison than any other convicted



Theodore Kaczynski, once a brilliant mathematician, became a recluse who mailed bombs to people. However, merely being “strange” or committing bizarre acts of violence does not qualify a person as legally insane.

person (Tanay, 1992). In short, the guilty but mentally ill verdict is the same as a guilty verdict.

Insanity pleas are offered in less than 1% of all criminal trials—and these are generally difficult, marginal cases. If the defendant is clearly out of touch with reality, the police and district attorney ordinarily bypass the trial and allow the defendant to enter a mental hospital. If the defendant has no serious signs of mental illness, the defense attorneys seldom attempt an insanity defense.

When the defendant's sanity is in question, the jury will listen to the views of psychologists and psychiatrists who are called as "expert witnesses." Expert witnesses often disagree with one another, however, thus leaving much to the judgment of a jury untrained in either the law or psychology. A jury that sympathizes with the defendant can use the insanity defense as a way of decreasing the penalty; a jury with less sympathy can refuse to find the defendant insane. As a citizen, voter, and potential jury member, you should be concerned about how to bring more rationality to this portion of our legal system.

CONCEPT CHECK

6. Someone who has been involuntarily committed to a mental hospital escapes and commits a murder. Will this person be judged not guilty by reason of insanity? (Check your answer on page 589.)

Predicting and Preventing Mental Illness

The traditional approach of psychotherapy has been to try to relieve psychological distress after it has developed. More recently, certain psychologists, especially community psychologists, began to pay more attention to prevention. Community psychologists focus on the needs of large groups rather than those of individuals. They distinguish between primary prevention, preventing a disorder from starting, and secondary prevention, identifying a disorder in its early stages and preventing it from becoming more serious.

Just as our society puts fluoride into drinking water to prevent tooth decay and immunizes people against contagious diseases, it can take action to prevent certain types of psychological disorders (Albee, 1986; Goldston, 1986; Long, 1986; Yoshikawa, 1994). For example, we could take the following actions:

- **Banning toxins.** The sale of lead-based paint has been banned because children who eat flakes of lead-based paint sustain brain damage. Other toxins in the air and water have yet to be controlled.
- **Educating pregnant women about prenatal care.** For example, women need to be informed that the use of alcohol or other drugs during pregnancy can cause brain dam-

age to the fetus and that certain bacterial and viral infections during pregnancy can impair fetal brain development and increase the risk of psychological disorders.

- **Jobs.** Helping people who lose their jobs to find new work enables them to regain their self-esteem.
- **Child care.** Providing better day-care facilities would contribute to the psychological health of both parents and children.
- **Improved educational opportunities.** Programs that get young people more interested in their schoolwork have made notable progress in decreasing juvenile delinquency.

These techniques are aimed at primary prevention for the entire community. Secondary prevention techniques can be targeted at specific individuals who are just beginning to show symptoms of a particular disorder. For example, psychologists identified fifth- and sixth-grade children prone to distress and then, for 12 weeks, taught them how to avoid negative beliefs about themselves. Those children showed fewer signs of depression over the next 2 years, compared to a similarly distressed group who did not receive this training (Gillham, Reivich, Jaycox, & Seligman, 1995).

Unfortunately, prevention turns out to be a difficult matter. For example, consider the problem of persuading people to change their behavior to help decrease the spread of AIDS (Helweg-Larsen & Collins, 1997).

THE MESSAGE The Science and Politics of Mental Illness

Suppose you are a storekeeper. Someone dressed as Batman stands outside your store every day shouting gibberish at anyone who comes by. Your once-thriving business draws fewer and fewer customers each day. The disturbing man outside does not seem to be breaking any laws, but you would like to have him removed from the area. Should he be taken to a mental hospital and treated, even though he insists that he is normal? It is a difficult decision. How disordered is he? What treatment would help him, and how much is it likely to help? And if nothing is done, what happens to your rights as a storekeeper?

Similarly, the insanity defense and all the other issues in this module are complicated issues that require political decisions by society as a whole, not just the opinions of psychologists or psychiatrists. Regardless of what career you enter, you will be a voter and potential juror, and therefore you will have as much to say about these issues as anyone else. The decisions deserve serious, informed consideration.

SUMMARY

- ★ **Mental hospitals and deinstitutionalization.** Today, very few patients spend a long time in mental hospitals. However, many states have released patients from mental



After screaming and stabbing himself in the throat with a screwdriver in front of the White House, this man was taken to a hospital for observation. Decisions about the treatment of such people must be based on the needs of both the individual and society.

hospitals without supplying adequate community mental health facilities. (page 584)

★ **Involuntary commitment.** Laws on involuntary commitment to mental hospitals vary. In some states, people can be committed only if they are dangerous; in other states, people can be committed if they are judged incompetent to make decisions about their own treatment. It is difficult to frame laws that ensure treatment for those who need it while also protecting the rights of those who have good reasons for refusing it. (page 585)

★ **Duty to warn.** The courts have ruled that a therapist who is convinced that a client is dangerous should warn the endangered person. Applying this rule in practice is difficult, however. (page 585)

★ **The insanity defense.** Some defendants accused of a crime are acquitted for reasons of insanity, which is a legal rather than a medical or psychological concept. The criteria for establishing insanity are vague and controversial. (page 586)

★ **Prevention of psychological disorders.** Psychologists and psychiatrists are increasingly concerned about preventing psychological disorders. Many preventive measures require the cooperation of society as a whole. (page 588)

Suggestions for Further Reading

- LaFond, J. Q., & Durham, M. L. (1992). *Back to the asylum*. New York: Oxford University Press. A history of the treatment of the mentally ill in the United States since about 1960, with a discussion of important controversies.
- Sheehan, S. (1982). *Is there no place on earth for me?* Boston: Houghton Mifflin. The story of a young woman with schizophrenia and her life in and out of mental hospitals.
- Simon, R. I. (1994). The law and psychiatry. In R. E. Hales, S. C.

Yudofsky, & J. A. Talbott (Eds.), *Textbook of psychiatry* (2nd ed.) (pp. 1297–1340). Washington, DC: American Psychiatric Press. A survey of many of the issues discussed in this module.

Terms

deinstitutionalization the removal of patients from mental hospitals (page 584)

Tarasoff the rule that a therapist who knew, or who should have known, that a client was dangerous is obligated to break the pledge of confidentiality and warn the endangered person (page 586)

M'Naghten rule the rule that a defendant is not criminally responsible if, at the time of committing an unlawful act, the person was laboring under such a defect of reason, from disease of the mind, as not to know the nature and quality of the act he was doing; or if he did know it, that he did not know he was doing wrong (page 587)

Durham rule the rule that a defendant is not criminally responsible if the activity was “a product of mental disease or defect” (page 587)

Model Penal Code the rule that a person is not responsible for criminal conduct if, at the time of such conduct and as a result of mental disease or defect, he or she lacks substantial capacity either to appreciate the criminality (wrongfulness) of this conduct or to conform his or her conduct to the requirements of law (page 587)

community psychologist a psychologist who focuses on the needs of large groups rather than those of individuals (page 588)

primary prevention preventing a disorder from starting (page 588)

secondary prevention identifying a disorder in its early stages and preventing it from becoming more serious (page 588)

Answer to Concept Check

- Not necessarily. Having a psychological disorder, even a severe one, does not automatically qualify a person as insane in the legal sense. The judge or jury must also find that the psychological disorder prevented the person from knowing what he or she was doing or made law-abiding behavior impossible. (page 588)

Web Resources

The Insanity Defense

www.psych.org/public_info/INSANI~1.HTM

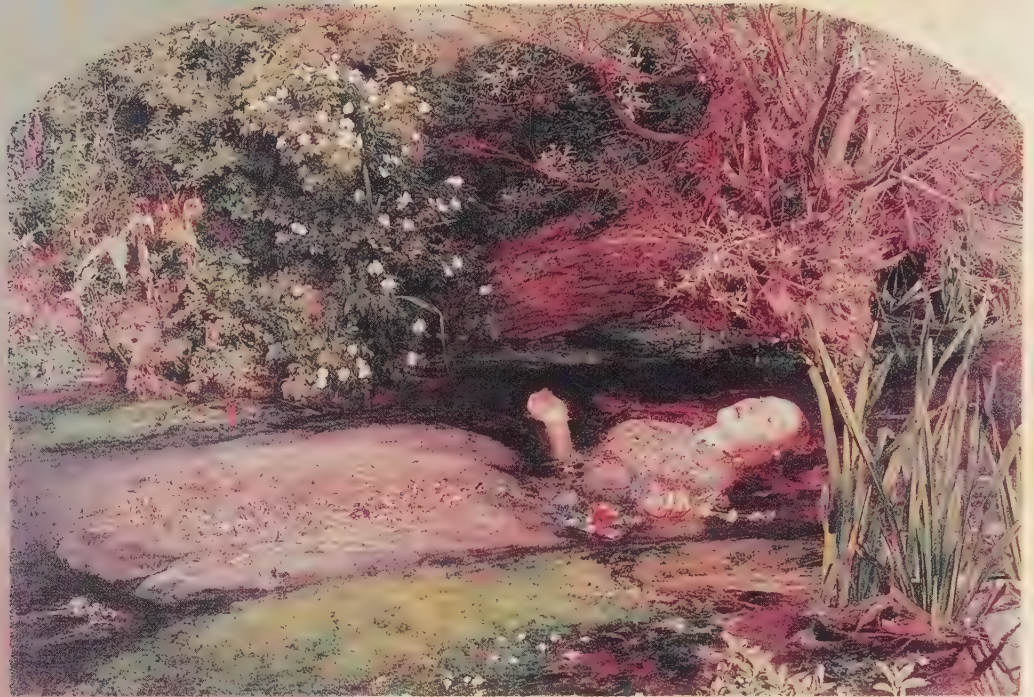
Information about the insanity defense from the American Psychiatric Association.

Landmark Cases in Forensic Psychiatry

bama.ua.edu/~jhooper/tableofc.html

In his excellent Forensic Psychiatry Resource Page, James Hooper offers brief summaries of almost 100 criminal cases that involve the insanity defense and civil cases that involve mental disorders. You can view the cases chronologically, alphabetically, or in a Java file.

Abnormalities and Therapies II: Explorations of Specific Disorders



Abnormal behavior can result from biological predisposition or stressful experience and, frequently, a combination of both influences. In Shakespeare's *Hamlet*, Ophelia became "mad" and drowned after Hamlet's behavior toward her changed from loving to cruel and he murdered her father.

16

★ MODULE 16.1

Anxiety and Avoidance Disorders

Disorders with Excessive Anxiety

Generalized Anxiety Disorder / Panic Disorder

Disorders with Exaggerated Avoidance Behaviors

Avoidance Learning and Its Persistence / Phobias /

LEARNING FEAR BY OBSERVATION: WHAT'S THE EVIDENCE? /

Drug Therapies for Phobias and Other Anxieties /

Obsessive-Compulsive Disorder / Therapies for

Obsessive-Compulsive Disorder

★ MODULE 16.2

Substance-Related Disorders

Substance Dependence (Addiction)

Predisposition to Addiction

Genetics and Family Background / WAYS OF PREDICTING

ALCOHOLISM: WHAT'S THE EVIDENCE?

Treatments for Alcoholism and Drug Addiction

*Treating Alcoholism / Specific Treatments for Opiate
Addiction*

★ MODULE 16.3

Mood Disorders

Depression

Genetic Predisposition to Depression / The Sex

Difference in the Prevalence of Depression / Events

That Precipitate Depression

Cognitive Aspects of Depression

The Learned-Helplessness Model / Explanatory Styles

Therapies for Depression

Cognitive Therapy / Biological Therapies for Depression /

The Advantages and Disadvantages of Antidepressant

Drugs / Electroconvulsive Shock Therapy

Seasonal Affective Disorder

Bipolar Disorder

A Self-Report: Depressive Phase / A Self-Report: Manic

Phase / Bipolar Cycles / Lithium Therapy for Bipolar

Disorder

Suicide

★ MODULE 16.4

Schizophrenia

The Symptoms of Schizophrenia

Hallucinations / Delusions / Disorders of Emotion and

Movement / The Thought Disorder of Schizophrenia /

The Distinction Between Positive and Negative

Symptoms

Types of Schizophrenia

Causes of Schizophrenia

Brain Damage / Genetics / The Neurodevelopmental

Hypothesis / Experience

Drug Therapies for Schizophrenia

Side Effects of Drug Therapies for Schizophrenia /

Family Therapy for Schizophrenia

In the 1700s and 1800s, medicine was very primitive by today's standards. Physicians would diagnose someone as being ill, but they rarely could distinguish one illness from another. Also, many of their treatments were the same for one illness as for another. They would recommend general all-purpose "tonics," suggest bed rest, sometimes apply leeches to withdraw blood, and so forth. Medical progress since then has been marked by an increasing ability to distinguish one disorder from another and to develop treatments that are useful for specific disorders.

Similarly, much of the progress in abnormal psychology has come from making appropriate distinctions between one disorder and another and developing treatments aimed at specific disorders. In this chapter, I shall concentrate on four major categories. As you read in Chapter 15, the most commonly diagnosed psychological disorders are anxiety disorders, substance abuse, and depression. The first three modules of this chapter cover those three disorders. The final module deals with schizophrenia, which is less common but in many cases extremely disabling.

Anxiety and Avoidance Disorders

Why do some people take extreme measures to avoid something that is harmless or only slightly dangerous?

Why do some people develop strange habits and rituals?

You go to the beach, looking forward to an afternoon of swimming and surfing. Would you still go in the water if someone tells you that a shark attacked two swimmers yesterday and has just been sighted close to shore? What if the shark attack occurred a month ago and no shark has been seen in the area since then? What if no shark has attacked anyone in this area, but someone saw a small shark there a few days ago? What if no shark has ever been seen near this beach, but recently you read a magazine story about shark attacks?

How much fear and caution is normal? Staying out of the water because you see a large shark is perfectly reasonable. Staying out of the water because you read an article about sharks is, by most people's standards, unreasonable. If you refuse even to look at photographs of the ocean because they might *remind* you of sharks, you have a serious problem indeed. Excessive fear and caution are linked to some of the most common psychological disorders.

Disorders with Excessive Anxiety

Many psychological disorders are marked by a combination of fear, anxiety, and attempts to avoid anxiety. Anxiety, unlike fear, is not generally associated with a specific situation. We feel fear in the presence of a hungry tiger, but our fear passes as soon as we get away. However, we cannot escape the anxiety we experience about dying or about our personal inadequacies. Some degree of anxiety is normal; anxiety becomes a problem only when it interferes with our ability to cope with everyday life.

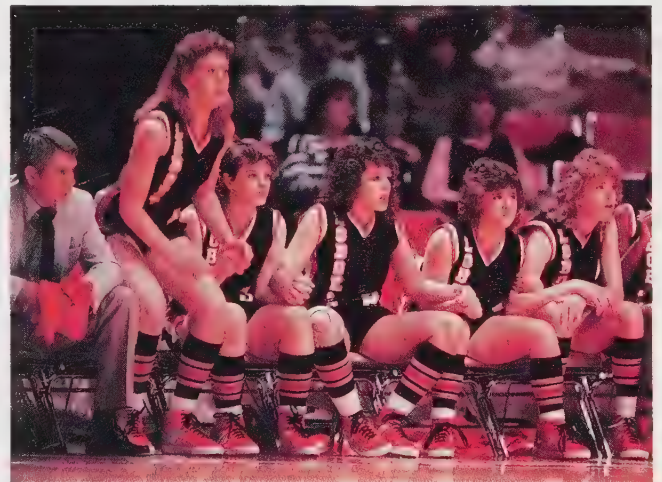
The people most prone to severe anxiety are those who feel helpless to control the major events of their lives (Chorpita & Barlow, 1998). Recall from Chapter 12 that shock, noise, and all sorts of other unpleasant events are more stressful for people who cannot predict or control them. Many people get a feeling of helplessness early in life and then develop a cognitive style of interpreting new events as uncontrollable. The result is vulnerability to anxiety.

Generalized Anxiety Disorder

People with generalized anxiety disorder are almost constantly plagued by exaggerated worries. They worry that "I might get sick," "My daughter might get sick," "I might lose my job," or "I might not be able to pay my bills." Although these people have no realistic reason for such worries—at least no more reason than anyone else—their worries persist and interfere with daily life. They grow tense, restless, irritable, and fatigued. About 5% of all people experience generalized anxiety disorder at some point in life, often in conjunction with depression, panic disorder, or other psychological problems (Wittchen, Zhao, Kessler, & Eaton, 1994).

Panic Disorder

People with panic disorder frequently have moderate anxiety and occasionally have attacks of sudden increased heart rate, chest pains, difficulty breathing, sweating, faintness, and trembling (see Figure 16.1). A panic attack usually lasts only a few minutes, although it can last an hour or more. During an attack, most people worry about fainting, having a heart attack, dying, or going crazy. Panic disorder occurs in 1–3% of adults at some time during their lives, in many cultures throughout the world, and it is more



Many situations evoke temporary anxiety or tension in almost anyone. Anxiety becomes a problem only if it is frequently more intense than the situation justifies.

FIGURE 16.1 People subject to panic attacks experience moderate anxiety almost constantly, as well as occasional periods of intense anxiety. During a panic attack, the sympathetic nervous system intensely activates the heart and other organs. People are generally frightened by this experience; some believe they are having a heart attack.



common in women than men (Weissman et al., 1997). Some people with panic disorder do in fact have *mitral valve prolapse*, an abnormality of heart structure that can cause heart flutter, chest pains, and breathing problems (Katendahl, 1993). However, most people with panic disorder have no apparent medical disorders.

One interpretation of panic disorder is that worried people *hyperventilate* (*breathe deeply and frequently*) to try to control their anxieties; in the process, they reduce their blood level of carbon dioxide. Many kinds of stressors increase the carbon dioxide level, so if the initial level is extremely low, even a small increase is a large percentage increase. The brain reacts to this sudden increase as if it were suffocating, thereby producing the full panic experience (Coplan et al., 1998; Klein, 1993). (See Figure 16.2.)

One treatment for panic disorders is to teach the person to avoid hyperventilating (Wolpe & Rowan, 1988).

Another is to teach the person to recognize sudden increases in heart rate and trembling as a sign of carbon dioxide fluctuations, and not as a sign of an impending heart attack.

Many people with panic disorder also have *social phobia*, *a severe avoidance of other people and an especially strong fear of doing anything in public*, where they might be embarrassed. Many also have *agoraphobia* (from *agora*, the Greek word for “marketplace”), *an excessive fear of open places or public places* (Magee, Eaton, Wittchen, McGonagle, & Kessler, 1996). Many relatives of people with panic disorder also have panic disorder, social phobia, or agoraphobia themselves (Goldstein et al., 1994; Kendler et al., 1995). Most psychologists believe that people with panic disorder develop their social phobia or agoraphobia because they are afraid of being incapacitated or embarrassed by a panic attack in a public place. In a sense,

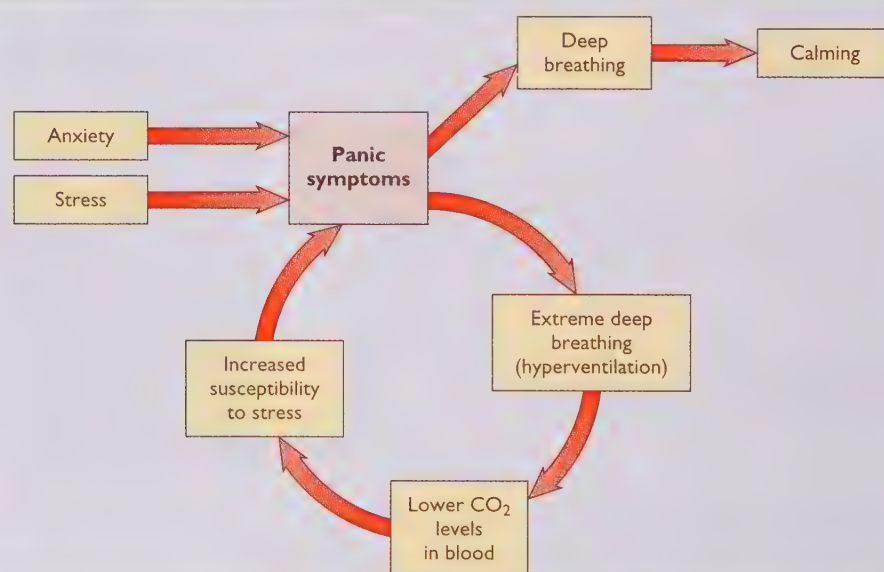


FIGURE 16.2 Deep breathing helps to calm a person, but hyperventilation increases the probability of another panic attack.

they are afraid of their own fear (McNally, 1990). To avoid the prospect of a public panic attack, they stay home as much as possible and rarely go out alone.

CONCEPT CHECK

1. Some psychologists advise people with panic attacks to stop worrying about their attacks and to adopt the attitude, “If it happens, it happens.” Why would they make this recommendation? (Check your answer on page 604.)

Disorders with Exaggerated Avoidance Behaviors

People learn to avoid punishment, as we saw in Chapter 6. In some cases, their efforts to avoid punishment become so extreme and persistent that they begin to interfere with daily activities. Let’s begin with some general observations on avoidance learning, which will be relevant to the later discussion of phobias (extreme fears) and compulsions (rituals designed to avoid unpleasant thoughts or events).

Avoidance Learning and Its Persistence

If you learn to do something for positive reinforcement, you will extinguish your response soon after you stop receiving reinforcements. Avoidance behaviors are different, however. Suppose you learn to press a lever to avoid electric shocks. Soon, you are responding steadily and you receive no shocks. Now, the experimenter disconnects the shock generator, without telling you. The extinction procedure has begun; you no longer need to press the lever. What will you do? You will continue pressing it, of course. As far as you can tell, nothing has changed; the response still “works.” Avoidance behaviors are highly resistant to extinction; once

someone learns a response to avoid mishap, the response continues long after it ceases to be necessary.

You can see how this tendency would support superstitions: Suppose you believe that Friday the 13th is a dangerous day. Every Friday the 13th, you are very cautious, but occasionally a misfortune happens anyway. The misfortune only confirms your belief that Friday the 13th is dangerous. The next Friday the 13th, nothing goes wrong. You conclude, “It helps to be careful on Friday the 13th. I was cautious all day long, so I avoided bad luck.” In other words, as long as you continue engaging in an avoidance behavior, you can never find out whether you are doing something useless.

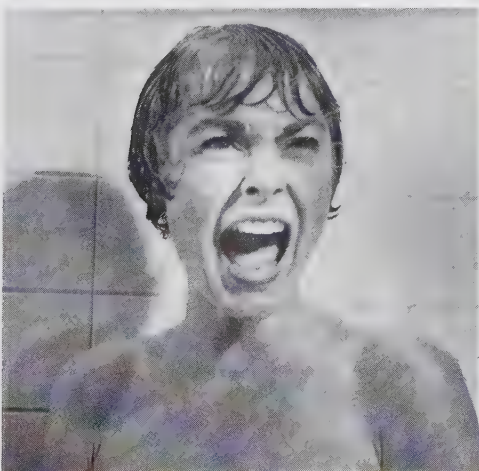
CONCEPT CHECK

2. Suppose you are an experimenter, and you have trained someone to press a lever to avoid shocks. Now, you disconnect the shock generator. Other than telling the person what you have done, what procedure could you use to facilitate extinction of the lever pressing? (Check your answer on page 604.)

Phobias

Terror is the only thing that comes close to how I feel when I think of moths. Their willowy, see-through wings always seem filthy. I remember being stuck in a car with a huge moth and my date, not knowing how terrified I was of moths, thought I was kidding when I told him I was afraid. It was terrible! I can feel it right now . . . the . . . feeling trapped and the moth with its ugly body flitting around so quickly, I couldn’t anticipate where it would go next. Finally that creature hit me in the arm and I screamed—it felt dirty and sleazy and then it hit me in the face and I began to scream uncontrollably. I had the terrible feeling it was going to fly into my mouth while I was screaming, but I couldn’t stop. (Duke & Nowicki, 1979, p. 244)

A phobia is best defined as a strong, persistent fear of a specific object, extreme enough to interfere with normal



Many people who watched the famous shower scene in the movie *Psycho* became frightened of taking showers. Actress Janet Leigh, who portrayed the woman killed in that shower scene, was herself so terrified that she has only rarely taken a shower since then. That is, fears and phobias can be learned, either from one’s own experiences or from vicarious experience.

living. It is often described as an unreasonable or irrational fear, but in most cases, the fear itself is not irrational. For example, a fear of snakes is not irrational, because snakes really can be dangerous. What is irrational is the excessive degree of the fear, which leads to avoidance behaviors—avoiding the object itself (such as a snake), places where the object might be lurking, or even anything that might remind the person of the feared object.

Confronting the object of a phobia can lead to sweating, trembling, and rapid breathing and heart rate. In most cases, people with phobias are not so much afraid of the object itself but of their own reactions (Beck & Emery, 1985). They fear that they will have a heart attack or that they will embarrass themselves by trembling or fainting. In many ways, phobias are like panic disorder, except that phobias are aroused by a specific object or event, whereas panic attacks occur at less predictable times.

Because most people with phobias are well aware that their fears are exaggerated, it does no good to tell them not to be afraid. In fact, attempts to reduce phobias by providing information sometimes backfire. One city tried to combat the phobia of elevators by posting signs on elevators throughout the city: “There is no reason to be afraid of elevators. There is almost no chance at all that the cable will break or that you will suffocate.” The signs actually increased the phobia of elevators.

The Prevalence of Phobias According to an extensive study of U.S. adults, about 11% of people suffer a phobia at some time in life, and 5–6% have a phobia at any given time (Magee et al., 1996). However, as with many other psychological disorders, phobias vary in degree from

mild to extreme, and the apparent prevalence of phobias increases or decreases, depending on how many borderline cases one includes. Figure 16.3 shows the most frequently reported phobias. Figure 16.4 shows the prevalence of phobias by age. Note the early onset and gradual decline in prevalence with age (Burke, Burke, Regier, & Rae, 1990).

The Learning of Phobias People seem to be born with a few fears, such as a fear of sudden loud noises, but most fears are learned, and so are most phobias. Indeed, some phobias can be traced to a specific event, such as when one child got locked in a trunk and developed a phobia of closed spaces. Another person developed a phobia of water by diving into a lake and discovering a corpse (Kendler et al., 1995).

John B. Watson, one of the founders of behaviorism, was the first to demonstrate the possibility of learning a fear (Watson & Rayner, 1920). Today, we would consider it unethical to try to create a fear, especially in humans, but in 1920, researchers felt less restraint. Watson and Rosalie Rayner studied an 11-month-old child, “Albert B.” who had previously shown no fear of white rats or other animals (Figure 16.5). They set a white rat down in front of Albert, and then, just behind him, they struck a large steel bar with a hammer. The sudden sound made Albert whimper and cover his face. After seven repetitions, the mere sight of the rat would make Albert cry and crawl away. Watson and Rayner declared that they had created a strong fear and that phobias in general might develop along similar lines.

Although Watson and Rayner’s study is open to serious methodological criticisms (Harris, 1979; Samelson, 1980), it led the way for later interpretations of phobias as learned

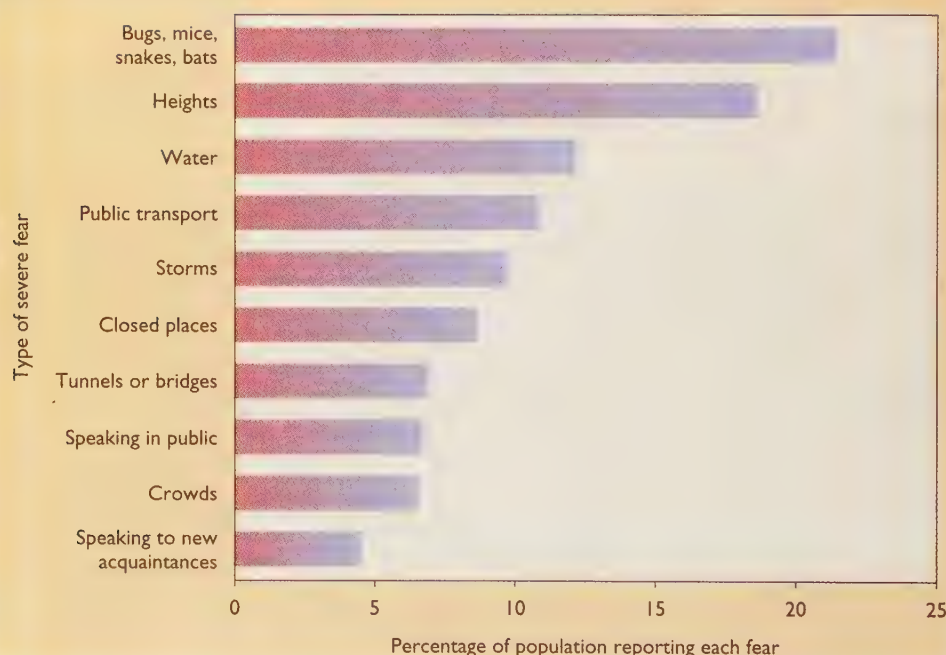


FIGURE 16.3 The most common phobias include phobias of snakes and other animals, heights, open places and crowds, and storms. Here, people reported their severe fears. Not all severe fears qualify as phobias, however.

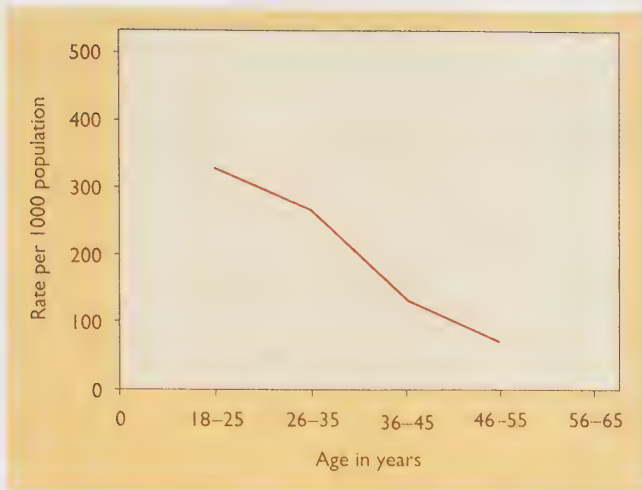


FIGURE 16.4 Most phobias are not lifelong conditions. Most young people with phobias lose their phobias by middle age, either spontaneously or because of therapy.

responses. Watson and Rayner's explanation of phobias failed to answer some important questions: Why do people develop phobias toward objects that have never injured them? Why are some phobias much more common than others? And why are phobias so persistent?

CONCEPT CHECK

3. In classical conditioning terms, what was the CS in Watson and Rayner's experiment? The UCS? The CR? The UCR? (Check your answers on page 604.)



FIGURE 16.5 John B. Watson argued that most fears, including phobias, are learned. Watson first demonstrated that this child, Little Albert, showed little fear of small animals. Then Watson paired the presentation of a white rat with a loud, frightening noise. Little Albert became afraid of the white rat; he also began showing fears of other small animals, odd-looking masks, and other objects he had not previously feared. (Courtesy of Professor Benjamin Harris.)

Learning Fear by Observation

Contrary to Watson and Rayner's explanation, almost half of all people with phobias have never had a painful experience with the object that they fear (Öst & Hugdahl, 1981). For example, many people have phobias, or at least intense fears, of snakes, spiders, or sharks, even though they have never been hurt by them. However, we have all heard about people who were injured or killed by snakes and sharks. As noted in the discussion of social learning in Chapter 6, we learn many things by watching or listening to others. Perhaps we learn our fears and phobias from other people.

That hypothesis is probably correct, but how can we demonstrate it? Susan Mineka and her colleagues demonstrated that monkeys learn fears by observing other monkeys (Mineka, 1987; Mineka, Davidson, Cook, & Keir, 1984). Her experiments show how animal studies can shed light on important human issues.

EXPERIMENT 1

Hypothesis Monkeys that have seen other monkeys show fear of a snake will develop such a fear themselves.

Method Monkeys that live in the wild generally develop a strong fear of snakes; however, monkeys reared in a laboratory show no fear of snakes. Mineka put a laboratory-reared monkey together with a wild-born monkey and let them both see a snake (Figure 16.6a). The lab monkey watched the wild monkey show signs of fear. Later, Mineka tested the lab monkey by itself to see whether it had acquired a fear of snakes.

Results When the lab monkey saw how frightened its partner was of the snake, it became frightened too (Figure 16.6b). It continued to be afraid of the snake when tested by itself, even months later.

Interpretation The lab monkey may have learned a fear of snakes because it saw that its partner was afraid of snakes. But Mineka considered another possible, though less likely, interpretation: The lab-reared monkey may have become fearful simply because it observed the other monkey's fear. That is, maybe it did not matter *what* the wild-reared monkey was afraid of. To test this possibility, Mineka conducted a second experiment.

EXPERIMENT 2

Hypothesis A monkey that sees another monkey show fear but does not know what the second monkey is afraid of will not develop the same fear itself.

Method A monkey reared in a lab watched a monkey reared in the wild through a plate-glass window. The wild monkey could look through another window, where it saw a snake. Thus, when the wild monkey shrieked and ran away

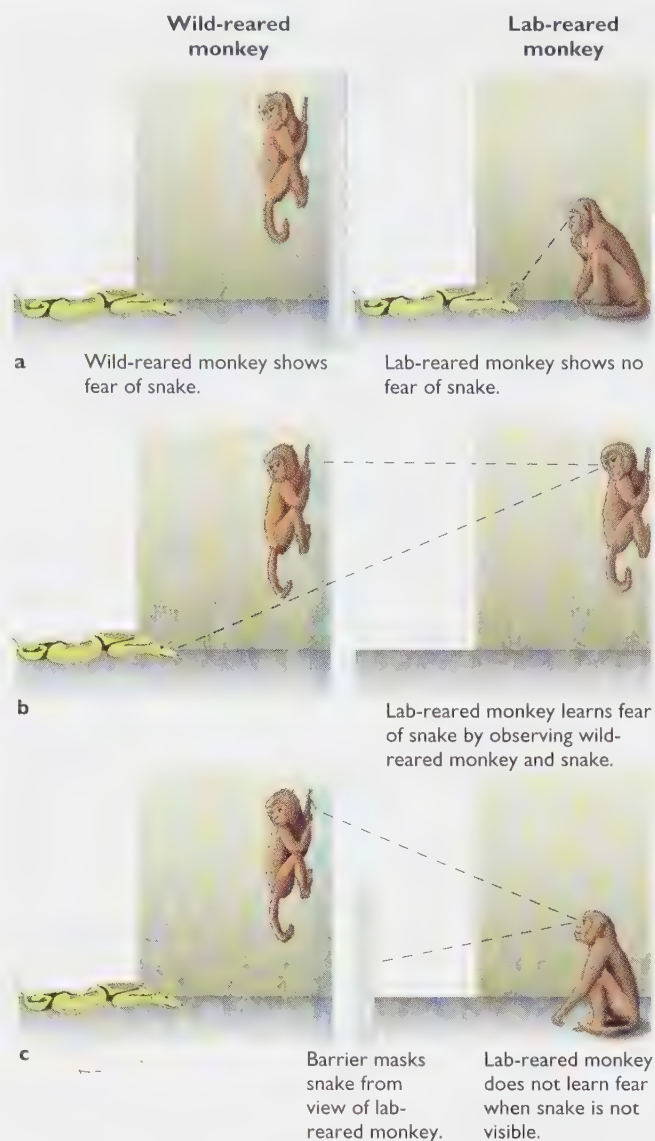


FIGURE 16.6 A laboratory-reared monkey learns to fear snakes from the reactions of a monkey reared in the wild. But if the snake is not visible, the lab-reared monkey fails to learn any fear.

from the snake, the lab monkey saw the wild monkey's fear but did not know what it was afraid of. Later, the lab monkey was placed close to a snake to see whether it would show fear.

Results The lab monkey showed no fear of the snake.

Interpretation To develop a fear of snakes, the observer monkey needed to see that the other monkey was frightened of snakes, not just that it was frightened (Figure 16.6c).

Note that, although the observer monkey had to see what the other monkey was afraid of, it did not have to see why it was afraid. Just seeing the other monkey's fear of the snake in Experiment 1 was enough. Humans not only observe other people's fears but can also tell one another what we are afraid of and why.



Why Some Phobias Are More Common Than Others

TRY IT YOURSELF

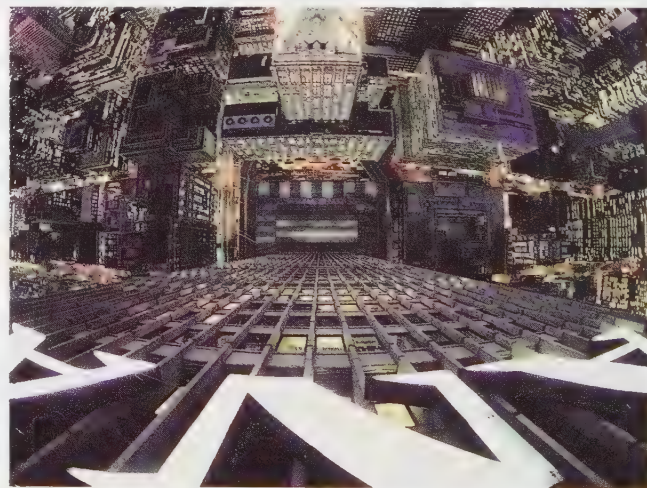
Imagine that you survey your friends. (You can actually survey them, if you like, but in this case it's pretty easy to imagine the results.) You ask them:

- Are you afraid of snakes?
- Are you afraid of cars?
- Have you ever been bitten by a snake or seen someone else get bitten by a snake?
- Have you ever been injured in a car accident or seen someone else get injured in a car accident?

I think you know what results to expect: A fair number of people will admit being afraid of snakes, some of them extremely afraid, even though very few have any firsthand experience with snakebites. Hardly anyone is afraid of cars, even though almost everyone has experienced or witnessed a car accident in which someone was injured. So, why do people develop some fears more readily than others?

The most common phobias are of open spaces, closed spaces, heights, lightning and thunder, animals, and illness. In contrast, few people have phobias of cars, guns, or tools—even though injuries from cars, guns, and tools are quite common. One explanation for this tendency is that, as Martin Seligman (1971) put it, people may be inherently “prepared” to learn certain phobias. For millions of years, people who quickly learned to avoid snakes, heights, and lightning have probably had a good chance to survive and to transmit their genes. We have not had enough time to evolve a tendency to fear cars and guns.

We have evidence to support this view from both monkey and human studies. Monkeys who watch a videotape of another monkey running away from a snake learn to fear snakes; monkeys who watch another monkey running away



Phobias are most likely to develop for dangerous events that we can neither predict nor control. They also develop for events that give us occasional unpleasant experiences without any safe or pleasant experiences. Can you therefore explain why so many people have phobias of falling from great heights (as in this photo) and so few people have phobias of cars?

from a flower show no fear of flowers (Mineka, 1987). People who receive electric shocks paired with pictures of snakes quickly develop a strong and persistent response to snake pictures; people who receive shocks paired with pictures of houses show a much weaker response (Öhman, Eriksson, & Olofsson, 1975).

There may be other reasons why some phobias are more common than others. One is that we have many safe experiences with cars and tools to outweigh any bad experiences. Most people have few safe experiences with snakes, spiders, or falling from high places. One study found that people who had extensive experiences with tarantula spiders lost their fears and sometimes even developed an interest in tarantulas as a hobby (Kleinknecht, 1982).

Another possible explanation is that people generally develop phobias of objects they cannot predict or control. If you are afraid of spiders, for example, you must be constantly on the alert for those tiny, unpredictable critters. Because you never know where they might be or when they might strike, you can never completely relax. Lightning is also unpredictable and uncontrollable. In contrast, electric outlets can also be dangerous, but you don't have to worry that they will take you by surprise. You must be on the alert for cars when you are near a road, but not at other times.

The Persistence of Phobias Well-established phobias can last a lifetime. Based on the discussion about avoidance learning, you can see why phobias are so difficult to extinguish: If you have learned to press a lever to avoid shock, you may not stop pressing long enough to find out that your response is no longer necessary. Similarly, if you stay away from snakes or heights or closed places because you have a phobia of them, you will never learn that your fear is exaggerated. In other words, you will not extinguish your fear.

Behavior Therapy for Phobias Recall the concept of behavior therapy from Chapter 15: A therapist sets a specific goal and applies learning techniques to help the client achieve that goal. One type of behavior therapy is particularly effective for phobias.

Phobias persist because people try to avoid the object of the phobia. The more effectively they avoid the object, the less chance they have to learn that their avoidance is unnecessary. That is, extinction does not occur. Therefore, the best way to extinguish a phobia is to expose a person to the object that arouses the fear. When nothing bad happens, the phobia fades.

One common and usually very successful treatment for phobia is systematic desensitization, a method of reducing fear by gradually exposing people to the object of their fear (Wolpe, 1961). Someone with a phobia of snakes, for example, is first given training in methods of relaxation. Then, the client is asked to lie on a comfortable couch with relaxing music playing in the background and with the therapist nearby. The therapist asks the client to imagine a small black-and-white photo of a snake. Next, he or she is asked

to imagine a full-color photo and, then, to imagine a real snake. After the client has successfully dealt with all of those images, the same sequence is repeated with real photos and eventually with a real snake (Figure 16.7).

The process resembles Skinner's shaping procedure (see Chapter 6): The person is given time to master each step before going on to the next. The client can say "stop," if the distress becomes threatening; the therapist then goes back several steps and repeats the sequence. Some people get through the whole procedure in a single 1-hour session; others need weekly sessions for a couple of months. Systematic desensitization can easily be combined with social learning: The person with a phobia watches other people who display a fearless response to the object.

Some therapists use a high-tech approach (Rothbaum et al., 1995): The client is equipped with a helmet that displays a virtual-reality scene, as shown in Figure 16.8. Then, without even leaving the office, the therapist can expose the client to the object of his or her fears. For example, a client with a phobia of heights can go up a glass elevator in a hotel, or walk across a narrow bridge over a chasm (Rothbaum et al., 1995). This virtual-reality technology gives the therapist excellent control of the situation, including even the option of turning off the display and removing the helmet, if the client becomes too fearful.

Flooding or implosion is a treatment where the person is exposed to the object of the phobia suddenly rather than gradually (Hogan & Kirchner, 1967; Rachman, 1969). (This treatment is called "flooding" because the patient is flooded with fear.) If you had a phobia of rats, for example, you might be told to imagine that you were locked in a room full of rats crawling all over you and viciously attacking you. The image arouses your sympathetic nervous system enormously, and your heart rate and breathing rate soar to high levels (Lande, 1982).



FIGURE 16.7 One of the most effective therapies for phobia is systematic desensitization: A therapist gradually exposes a client to the object of the phobia, first in imagination and later in reality. A similar procedure is exposure therapy; the therapist demonstrates a lack of fear of the object and encourages the client to do the same.

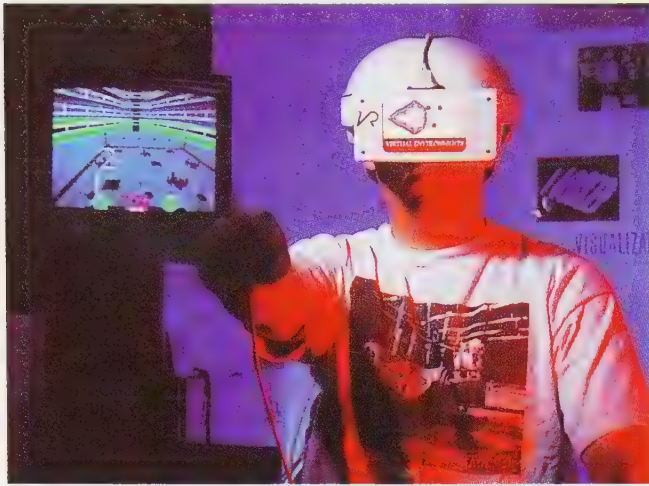


FIGURE 16.8 Some therapists provide a virtual-reality environment, so that a patient with a phobia of heights can experience heights under carefully controlled conditions.

The human sympathetic nervous system is not capable of maintaining extreme arousal for very long, however, so within a few minutes, your heart rate and breathing rate begin to decline. A little later, you report that you feel more relaxed, even though the therapist continues to suggest gory images of what rats are doing to you. Once you have reached this point, the battle is half won. The main fear of people with phobias is actually that their own fear will be overwhelming or that it will lead to a heart attack; when they learn that they can withstand even the most frightening experience, they become less afraid of their own responses.

Drug Therapies for Phobias and Other Anxieties

Tranquilizers are among the most widely prescribed drugs in the United States. The most common tranquilizers are the benzodiazepines (BEN-zo-die-AZ-uh-peens), including drugs with the trade names Valium, Librium, and Xanax. Benzodiazepines relieve anxiety, relax the muscles, induce sleep, and

inhibit epileptic seizures. Sometimes, they are prescribed for people with panic disorder, phobia, or other anxiety disorders; they are also frequently prescribed for people with no diagnosable disorder. These drugs can serve as a means of relieving an occasional episode of anxiety or as a sleeping pill. One disadvantage is that they can be habit-forming; another is that they suppress symptoms only temporarily. For example, someone who takes pills to combat panic disorder is likely to resume having panic attacks after discontinuing the pills (Wiborg & Dahl, 1996). In many cases, therapists treat anxiety disorders more effectively with antidepressant drugs, which we shall discuss later in this chapter.

Brain researchers now understand in reasonable detail how benzodiazepines work. The neurotransmitter GABA stimulates receptors on a neuron's membrane, as shown in Figure 16.9. On its own, GABA attaches moderately easily to this receptor, opening a channel for chloride ions (Cl^-) to enter the neuron. Benzodiazepines alter the shape of the receptor complex to enable GABA molecules to attach more readily. Thus, benzodiazepines indirectly facilitate transmission at these synapses (Macdonald, Weddell, & Gross, 1986). Facilitation of these synapses decreases anxiety and promotes sleep.

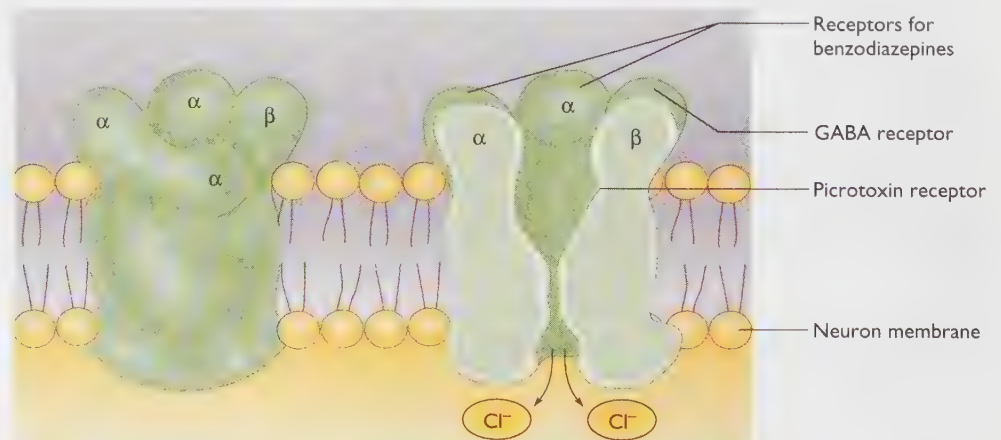
CONCEPT CHECKS

4. Alcohol facilitates transmission at GABA synapses. What effect should we expect if someone took both alcohol and a benzodiazepine tranquilizer?
5. How does systematic desensitization resemble extinction of a learned shock-avoidance response?
6. How is the flooding procedure related to the James-Lange theory of emotions that was discussed in Chapter 12? (Check your answers on page 604.)

Obsessive-Compulsive Disorder

People with obsessive-compulsive disorder have two kinds of problems: An obsession is a repetitive, unwelcome stream of thought. For example, such people might

FIGURE 16.9 Benzodiazepine tranquilizers attach to a site on the receptor complex for the neurotransmitter GABA. When they attach, they modify the shape of the receptor, making it easier for GABA to stimulate its receptor. Thus the benzodiazepines indirectly facilitate transmission at this synapse, thus enabling chloride molecules to enter the neuron. The effect on behavior is to relieve anxiety. (Based on Guidotti et al., 1986.)



find themselves constantly imagining gruesome scenes, worrying that they are about to kill someone, dwelling on doubts about their religion, or thinking “I hate my sister, I hate my sister.” The harder they try to escape such thoughts, the more repetitive they become. A **compulsion** is a repetitive, almost irresistible action. Obsessions generally lead to compulsions, as an itching sensation leads to scratching.

The Persistence of Obsessions People with obsessive-compulsive disorder feel a combination of guilt and anxiety over certain thoughts or impulses. They feel persistent, frightening impulses—perhaps an impulse to engage in a sexual act that they consider shameful, an impulse to hurt someone they love, or an impulse to commit suicide. They decide, “Oh, what a terrible thing to think. I don’t want to think such a thing ever again.” And so they resolve to shut the thought or impulse out of their consciousness. However, the harder one tries to block out a thought, the more intrusive it becomes. As a child, the Russian novelist Leo Tolstoy once organized a club with a most unusual qualification for membership: A prospective member had to stand alone in a corner *without thinking about a white bear* (Simmons, 1949).

**TRY IT
YOUR-
SELF**

If you think that sounds easy, try it. Normally, you probably go months at a time between thoughts about polar bears, but when you try *not* to think about them, you can think of little else.

In one experiment, college students were asked to tape-record everything that came to mind during 5 minutes but to try *not* to think about white bears. If they did, they were to mention it and ring a bell. Subjects reported thinking about bears a mean of more than six times during the 5 minutes (Wegner, Schneider, Carter, & White, 1987). Afterward, they reported that almost everything in the room reminded them of white bears. Evidently, attempts to suppress a thought are likely to backfire, even with an emotionally trivial thought about something such as white bears; you can imagine how hard it must be with severely upsetting thoughts.

One Type of Compulsion: Cleaning People with obsessive-compulsive disorder have any of several kinds of compulsions. Some collect things. (One man collected



Most people seldom think about polar bears. But try to avoid thinking about them and you can think of little else but polar bears. Similarly, many attempts to block obsessive thinking backfire.

newspapers under his bed until they raised the bed so high it almost touched the ceiling.) Others have odd habits, such as touching everything they see, trying to arrange objects in a completely symmetrical manner, or walking back and forth through a doorway nine times before leaving a building. One obsessive-compulsive person could not go to sleep at night until he had counted the corners of every object in the room to make sure that the total number of corners was evenly divisible by 16. If it was not, he would add or remove objects from the room until the room had an acceptable total of corners. (The Obsessive-Compulsive Foundation produces a button that reads, “Every Member Counts”!)

The most common compulsions are cleaning and checking. Obsessive-compulsive cleaning is similar to a phobia of dirt. Here is a description of a severe cleaning compulsion (Nagera, 1976):

“R,” a 12-year-old boy, had a long-standing habit of prolonged bathing and hand washing, dating from a film about germs he had seen in the second grade. At about age 12, he started to complain about “being dirty” and having “bad thoughts,” but he would not elaborate. His hand washing and bathing became longer and more frequent. When he bathed, he carefully washed himself with soap and washcloth all over, including the inside of his mouth and the inside of each nostril. He even opened his eyes in the soapy water and carefully washed his eyeballs. The only part he did not wash was his penis, which he covered with a washcloth as soon as he entered the tub.

Coupled with R.’s strange bathing habits, he developed some original superstitions. Whenever he did anything with one hand, he immediately did the same thing with the other hand. Whenever anyone mentioned a member of R.’s family, he would mention the corresponding member of the other person’s family. He always walked to school by the same route, being careful never to step on any spot he had ever stepped on before. (After a while, this became a serious strain on his memory.) At school, he would wipe the palm of his hand on his pants after any “good” thought; at home, he would wipe his hand on his pants after any “bad” thought.

R.’s problems were traced to a single event. Just before the onset of his exaggerated behaviors, R. and another boy had pulled down their pants and looked at each other. Afterward, he felt guilty and full of anxiety that he might do the same thing again. The constant bathing was apparently an attempt to wash away his feelings of “dirtiness.” The superstitious rituals were an attempt to impose rigid self-control. His underlying reasoning could be described as: “If I can keep myself under perfect control at all times, even following these rigid and pointless rules, I will never again lose control and do something shameful.”

Another Type of Compulsion: Checking An obsessive-compulsive checker “double-checks” everything. Before going to bed at night, he or she checks to make sure that all the doors and windows are locked and that all the water faucets and gas outlets are turned off. But then the

TABLE 16.1 Obsessive-Compulsive Cleaners and Checkers

	CLEANERS	CHECKERS
Sex distribution	Mostly female	About equally male and female
Dominant emotion	Anxiety, similar to phobia	Guilt, shame
Speed of onset	Usually rapid	More often gradual
Life disruption	Dominates life	Usually does not disrupt job and family life
Ritual length	Less than one hour at a time	Some go on indefinitely
Feel better after rituals?	Yes	Usually not

SOURCE: Rachman & Hodgson, 1980.

question arises, “Did I *really* check them all, or did I only imagine it?” So everything has to be checked again. And again. “After all, I may accidentally have unlocked one of the doors when I was checking it.” Obsessive-compulsive checkers can never decide when to stop checking; they may go on for hours and even then not be satisfied.

Obsessive-compulsive checkers have been known to check every door they pass to see whether anyone has been locked in, to check trash containers and bushes to see whether anyone has abandoned a baby, to call the police every day to ask whether they have committed a crime that they have forgotten, and to drive back and forth along a street to see whether they ran over any pedestrians the last time through (Pollak, 1979; Rachman & Hodgson, 1980).

Why do “checkers” go on checking? According to some reports, they do not trust their memory of what they have done (Sher, Frost, Kushner, Crews, & Alexander, 1989). In one study, the experimenters asked several people—among them some obsessive-compulsive checkers—to read a list of words and to think their opposites. (For instance, when they saw “NORTH-S . . .,” they would think *south*.) Then the experimenters combined the two sets of words—the words on the list and their opposites—and asked the subjects to identify which words they had read and which ones they had only thought. The obsessive-compulsive checkers remembered about as accurately as the control group did, but their confi-

dence in their answers was significantly lower. Compared to the control group, the checkers were less confident of their ability to distinguish between what had actually happened and what they had just imagined (Sher, Frost, & Otto, 1983).

Table 16.1 summarizes key differences between obsessive-compulsive cleaners and checkers. Table 16.2 lists some items from a questionnaire on obsessive-compulsive tendencies (Rachman & Hodgson, 1980). Try answering these questions yourself, or try guessing how an obsessive-compulsive person would answer them. (The few items listed here are not sufficient to diagnose someone as obsessive-compulsive. So don’t obsess about it if you agreed with all the obsessive-compulsive answers.)

**TRY IT
YOUR-
SELF**

Possible Predispositions or Causes About 2–3% of all people in the United States suffer from obsessive-compulsive disorder at some time during their life (Karno, Golding, Sorenson, & Burnam, 1988). The disorder occurs most frequently among hardworking, perfectionistic people of average or above-average intelligence. It can develop either suddenly or gradually, usually beginning between the ages of 10 and 25. Nearly all people with obsessive-compulsive disorder have some insight into their own behavior and realize that their rituals are inappropriate. However, that realization does not stop the rituals.

TABLE 16.2 Obsessive-Compulsive Tendencies

1. I avoid public telephones because of possible contamination.	T	F
2. I frequently get nasty thoughts and have difficulty getting rid of them.	T	F
3. I usually have serious thoughts about the simple everyday things I do.	T	F
4. Neither of my parents was very strict during my childhood.	T	F
5. I do not take a long time to dress in the morning.	T	F
6. One of my major problems is that I pay too much attention to detail.	T	F
7. I do not stick to a very strict routine when doing ordinary things.	T	F
8. I do not usually count when doing a routine task.	T	F
9. Even when I do something very carefully, I often feel that it is not quite right.	T	F

SOURCE: Rachman & Hodgson, 1980. (Check typical answers on page 604.)

Obsessive-compulsive disorder tends to run in families, suggesting a possible genetic basis. Many of the relatives who do not have obsessive-compulsive disorder have other anxiety disorders (Black, Noyes, Goldstein, & Blum, 1992).

Therapies for Obsessive-Compulsive Disorder

Obsessive-compulsive disorder can be treated through various forms of psychotherapy, including *exposure therapy*, which resembles systematic desensitization for phobias: The obsessive-compulsive person is exposed to the situation in which he or she ordinarily performs certain rituals and is then prevented from performing them. For example, an obsessive-compulsive cleaner would be prevented from cleaning, or someone who ordinarily spends hours checking the doors and windows before going to sleep would be prevented from making the rounds. The point is to demonstrate that nothing catastrophic occurs if one leaves a little mess in the house or runs a slight risk of leaving a door unlocked.

The drug clomipramine (Anafranil) and related drugs are also helpful for at least half of obsessive-compulsive patients (Greist, Jefferson, Kobak, Katelnick, & Serlin, 1995). Clomipramine prolongs the effects of the neurotransmitter serotonin by preventing the presynaptic neuron from reabsorbing it. That is, after an axon releases serotonin from its terminal button, clomipramine causes the serotonin to remain longer than usual in the synapse, reexciting the postsynaptic neuron. Researchers do not yet understand why prolonging serotonin activity should relieve obsessive-compulsive disorder.

THE MESSAGE Anxieties and Avoidance

Phobia and obsessive-compulsive disorder illustrate some of the possible links between emotions and cognitions. At the risk of seriously oversimplifying, we could say that people with phobias experience emotional attacks because of their cognitions about a particular object, whereas people with obsessive-compulsive disorder experience repetitive cognitions for emotional reasons. In both conditions, most people are cognitively aware that their reactions are exaggerated, but mere awareness of the problem does not correct it. Dealing with such conditions requires attention to emotions, cognitions, and the links between them.

S U M M A R Y

★ **Anxiety disorder and panic disorder.** People with generalized anxiety disorder or panic disorder experience extreme anxiety. Panic disorder is characterized by episodes of disabling anxiety, some of which may be triggered by hyperventilation. (page 593)

★ **Persistence of avoidance behaviors.** Once an individual has learned a shock-avoidance response, the response can persist long after the possibility of shock has been removed. As with shock-avoidance responses, phobias and obsessive-compulsive disorder persist because people do not discover that their avoidance behaviors are unnecessary. (page 595)

★ **Phobia.** A phobia is a fear so extreme that it interferes with normal living. Phobias are learned through observation as well as through experience. (page 595)

★ **Common phobias.** People are more likely to develop phobias of certain objects than of others; for example, snake phobias are more common than car phobias. The objects of the most common phobias have menaced humans throughout evolutionary history. They pose dangers that are difficult to predict or control, and they are generally objects with which we have had few safe experiences. (page 598)

★ **Systematic desensitization of phobias.** A common therapy for phobia is systematic desensitization. The patient is taught to relax and is then gradually exposed to the object of the phobia. Flooding is similar, except that the person is exposed to the object suddenly. (page 599)

★ **Tranquilizers.** Drugs that facilitate activity at GABA synapses help to relieve anxiety. (page 600)

★ **Obsessive-compulsive disorder.** People with obsessive-compulsive disorder try to avoid certain thoughts or impulses that cause anxiety or guilt. They also have repetitive behaviors. (page 600)

★ **Types of obsessive-compulsive disorder.** Two common types of compulsion are cleaning and checking. Cleaners try to avoid any type of contamination. Checkers constantly double-check themselves and invent elaborate rituals. (page 601)

Suggestion for Further Reading

de Silva, P., & Rachman, S. (1998). *Obsessive-compulsive disorder: The facts* (2nd ed.). Oxford, England: Oxford University Press. Concise overview of the symptoms and treatments for obsessive-compulsive disorder.

Terms

generalized anxiety disorder a disorder in which people are almost constantly plagued by exaggerated worries (page 593)

panic disorder a disorder characterized by frequent bouts of moderate anxiety and occasional attacks of sudden increased heart rate, chest pains, difficulty breathing, sweating, faintness, and trembling (page 593)

hyperventilate breathe deeply and frequently (page 594)

social phobia a severe avoidance of other people and an especially strong fear of doing anything in public (page 594)

agoraphobia an excessive fear of open places or public places (page 594)

phobia a strong, persistent fear of a specific object, extreme enough to interfere with normal living (page 595)

systematic desensitization a method of reducing fear by gradually exposing people to the object of their fear (page 599)

flooding or implosion a treatment for phobia in which the person is suddenly exposed to the object of the phobia (page 599)

obsessive-compulsive disorder a condition with repetitive thoughts and actions (page 600)

obsession a repetitive, unwelcome stream of thought (page 600)

compulsion a repetitive, almost irresistible action (page 601)

Answers to Concept Checks

1. Worrying about anything—even panic attacks themselves—often prompts these people to hyperventilate, and hyperventilation can in turn lead to another panic attack. (page 595)
2. Temporarily prevent the person from pressing the lever. Only by ceasing to press it does the person discover that pressing is not necessary. (page 595)
3. The CS was the white rat. The UCS was the loud noise. The CR and the UCR were a combination of crying and other reactions of fear. (page 597)
4. The combined effect of alcohol and a benzodiazepine tranquilizer would decrease anxiety, relax the muscles, and induce sleep more effectively than either alcohol or a benzodiazepine could by itself. In fact, the combined effect can be so strong as to be dangerous, even fatal. People given prescriptions for benzodiazepine tranquilizers are warned not to take them in conjunction with alcohol. (page 600)
5. The method of extinguishing a learned shock-avoidance response is to prevent the response so that the individual learns that the failure to respond is not followed by shock. Similarly, with systematic desensitization, the patient is prevented from fleeing the feared stimulus; he or she therefore learns that the danger is not as great as imagined. (page 600)

6. The flooding procedure is compatible with the James-Lange theory of emotions, which holds that emotions follow from perceptions of body arousal. In flooding, as arousal of the autonomic nervous system decreases, the person perceives: “I am calming down. I must not be as frightened of this situation as I thought I was.” (page 600)

Answers to Other Questions in the Text

Typical answers for obsessive-compulsive people (page 602):

1. T; 2. T; 3. T; 4. F; 5. F; 6. T; 7. F; 8. F; 9. T

Web Resources

Anxiety Disorders

www.psych.org/public_info/anxiety.html

Panic Disorder

www.psych.org/public_info/panic.html

Phobias

www.psych.org/public_info/phobias.html

Post Traumatic Stress Disorder

www.psych.org/public_info/ptsd.html

Obsessive-Compulsive Disorder

www.psych.org/public_info/PDF/ocd.pdf

Each American Psychiatric Association's *Let's Talk Facts* pamphlet describes the disorder, the major symptoms, what we know about the causes, and treatment. (OCD is available only in PDF format. You will need the free Adobe Acrobat Reader, available from www.adobe.com/prodindex/acrobat/readstep.html, to read or print it.)

Substance-Related Disorders

Why do people sometimes abuse alcohol and other drugs?

What can be done to help people quit?

How would you like to volunteer for a little experiment? I want to implant a device in your head to control your brain activity—something that will automatically lift your mood. There are still a few kinks in it, but most of the people who have tried it say that it makes them feel good at least some of the time, and some people say it makes them feel very happy.

I should tell you about the possible risks: My device will endanger your health and will reduce your life expectancy by, oh, 10 years or so. Some people believe it may cause permanent brain damage, but they haven't proved that charge, so I don't think you should worry about it. Your behavior will change a good bit, though. You may have difficulty concentrating, for example. The device affects some

people more than others. If you happen to be one of those who are strongly affected, you will have difficulty completing your education, getting or keeping a job, and carrying on a satisfactory personal life. But if you are lucky, you can avoid all that. Anyway, you can quit the experiment any time you want to. You should know, though, that the longer the device remains in your brain, the harder it is to get it out.

I cannot pay you for taking part in this experiment. In fact, *you* will have to pay *me*. But I'll give you a bargain rate: only \$5 for the first week and then a little more each week, as time passes. One other thing: Technically speaking, this experiment is illegal. We probably won't get caught, but if we do, we could both go to jail.

What do you say? Is it a deal? I presume you will say "no." I get very few volunteers. And yet, if I change the term *brain device* to *drug* and change *experimenter* to *drug peddler*, it is amazing how many volunteers come forward.

For some people, using alcohol or drugs is apparently a harmless pleasure. For others, it is extremely destructive. In Chapter 5, we examined the effects of several drugs on behavior. Instead of reviewing all of those drugs again here, we shall focus on substance abuse, principally of alcohol and opiates—addictions that have been familiar to humans for centuries and which continue to be major problems today.

Substance Dependence (Addiction)

Some people drink alcohol or experiment with other drugs only in moderation. Others continue until they jeopardize their health, their work or their education, and the welfare of their family. Substance abusers do not all follow the same pattern; they may use the substance daily, only on weekends, or only during sporadic binges. Those who find it difficult or impossible to quit a self-destructive habit are said to have a **dependence on a substance or addiction to it**. It is difficult to draw broad generalizations about the behavior of people who have a substance dependence. Many people with a dependence ask themselves, "am I an alcoholic?" or "am I addicted?" and manage, for a time, to convince themselves that they are not, because they do not use the substance every day or because they are able to keep a job. However, alcoholism and addiction are not simple yes-or-no matters. They vary in degree from mild to severe, and people react in different ways to drugs (Miller & Brown, 1997). It is possible to be an alcoholic or a drug addict and still keep a job; it is also possible to go days or weeks without using the substance at all. The questions to ask are, "Does the substance cause troubles in my life, and do I often take more than I had decided I would?" If the answers are "yes," then you have a problem.

Almost any substance can be addictive under certain circumstances. In one hospital ward where alcoholics were being treated, one of the patients moved his bed into the men's room (Cummings, 1979). At first, the hospital staff



Addiction is in the user, not in the drug. Some people will continue using drugs even though they know the drugs endanger their health, limit their opportunities, and provide them with little pleasure.

ignored this curious behavior. Then, one by one, other patients moved their beds into the men's room. Eventually, the staff discovered what was going on. These men, deprived of alcohol, had discovered that they could get "high" by drinking enormous amounts of water! By drinking about 30 liters (7.5 gallons) of water per day and urinating the same amount (which was why they moved into the men's room), they managed to alter the acid-to-base balance of their blood enough to produce something like drunkenness. Is water addictive? Evidently it can be, for some people.

Nevertheless, some substances are much more likely than others to be addictive. Nearly all of the commonly addictive drugs stimulate types D_2 , D_3 , and D_4 dopamine receptors in the brain (Koob & LeMoal, 1997). As the drugs stimulate these receptors, they produce reinforcing effects, but they also decrease the sensitivity of the receptors. Someone who quits a drug after using it extensively experiences an intense period of withdrawal when the receptors are not stimulated by the drugs and not very responsive to normal events, either. Withdrawal symptoms can be quite unpleasant. Long after the symptoms are gone, however, the person can still experience cravings for the drug (Pisasecki, Kenford, Smith, Fiore, & Baker, 1997). One speculation is that the dopamine receptors are still somewhat unresponsive, so the person gets little reinforcement from everyday events (Nestler & Aghajanian, 1997; Pidoplichko, De Biasi, Williams, & Dani, 1997).

Other things being equal, the more rapidly a substance enters the brain, the more addictive it is likely to be. Cigarettes are more addictive than cigars, for example, because smokers inhale cigarette smoke more deeply, thus allowing the nicotine to enter the bloodstream and reach the brain more quickly (Bennett, 1980). For the same reason, crack cocaine is more addictive than other forms of cocaine.

You have no doubt learned that the cigarette smoking habit is based largely on nicotine addiction. One type of evidence supporting this conclusion is that people find it easier to quit smoking cigarettes if they have a replacement source of nicotine, such as a nicotine patch, nicotine chewing gum, or nicotine nasal spray (Cepeda-Benito, 1993; Schneider, 1994). "If that is so," you may have wondered, "why have so many smokers switched to low-tar, low-nicotine cigarette brands? Wouldn't those brands fail to satisfy a nicotine craving?" Yes, they would . . . if they delivered low nicotine! A low-tar, low-nicotine cigarette has the same kind of tobacco as other cigarettes, but a different filter. The special feature of the filter is a row of little air holes, as shown at the top of Figure 16.10. The theory is that air entering through these air holes will dilute the tobacco smoke coming through the barrel of the cigarette. People who smoke that way do indeed get lower tar and nicotine. However, many smokers wrap their fingers around the air holes, either accidentally or intentionally. Some even wrap tape over the holes. Regardless of whether people cover the holes, those who switch to low-nicotine cigarettes generally inhale more deeply than they did when smoking regular cigarettes, and they smoke more cigarettes per day. As a result



FIGURE 16.10 "Low-nicotine" cigarettes have a row of small holes in the filter; room air is supposed to enter through those holes and therefore dilute the tobacco smoke. If people smoke such cigarettes without covering the air holes, little tar and nicotine passes through the cigarette, as we see from the relatively clean filter tip. However, if people cover the holes with tape or (more commonly) with their fingers, they will receive about as much tar and nicotine as they would from any other filtered cigarette.

of all these changes in their smoking behavior, people smoking low-nicotine cigarettes inhale about as much tar and nicotine per day as do people smoking regular cigarettes (Benowitz, 1986; Kozlowski, Frecker, Khouw, & Pope, 1980).

SOMETHING TO THINK ABOUT

Are any addictive behaviors beneficial? (If a behavior is beneficial, can we call it *addictive*?) ★

CONCEPT CHECK

7. Methadone, an opiate drug, can be taken either by injection (entering the blood rapidly) or as a pill (entering the blood slowly). Which route is more likely to lead to addiction? (Check your answer on page 613).

Predisposition to Addiction

If, when you honestly want to, you find you cannot quit entirely, or if when drinking, you have little control over the amount you take, you are probably alcoholic.

—ANONYMOUS (1955)

Why do some people become alcoholics, whereas others drink only in moderation? If we could identify the probable alcoholics early enough, we might be more effective in helping them either abstain or drink in moderation. At least, psychologists would like to try; clearly, if we wait until someone has a serious alcohol problem, fighting the problem is difficult and frustrating.

Genetics and Family Background

Alcoholism, the habitual overuse of alcohol, is the most common and the most costly form of drug abuse in the United States and Europe. An estimated 25–40% of all hospital patients suffer from complications caused by alcohol abuse (Holden, 1987), and a great many people who seek help for depression, anxiety, or other psychological disorders suffer from alcoholism or other substance abuse as well.

There is convincing evidence that genetics can cause a predisposition to alcoholism, although different studies give conflicting results about the importance of that role (Cadoret, Troughton, & Woodworth, 1994). Biological children of alcoholic parents have an increased risk of alcoholism, even if they are adopted by nonalcoholics (Cloninger, Bohman, & Sigvardsson, 1981; Vaillant & Milofsky, 1982). However, no one is totally safe; even people with no family history of alcoholism sometimes develop an alcohol problem. Researchers distinguish between Type I and Type II alcoholism, also known as Types A and B (Brown, Babor, Litt, & Kranzler, 1994; Devor, Abell, Hoffman, Tabakoff, & Cloninger, 1994): **Type I (or Type A) alcoholism is generally less severe, occurs about equally in men and women, has less dependence on genetics, and develops gradually, presumably in response to the difficulties of life. Type II (or Type B) alcoholism is generally more severe, occurs mostly in men, is more often associated with aggressive behavior or antisocial personality, shows a strong genetic tendency, and usually begins early in life.** Table 16.3 summarizes this distinction. Naturally, not every alcoholic fits neatly into one category or the other, but the distinction accounts for most cases.

The incidence of alcoholism is greater than average among people who grew up in families marked by conflict



Jewish and Italian cultures, which stress moderation, have a fair amount of alcohol use but relatively little alcohol abuse.

between parents, poor relationships between parents and children, and inadequate parental supervision of the children (Schulsinger, Knop, Goodwin, Teasdale, & Mikkelsen, 1986; Zucker & Gomberg, 1986). People who have had extremely traumatic experiences are especially likely to turn to alcohol, presumably as a means of escape (Stewart, 1996). Among college students, alcohol abuse is most common among those who are having academic troubles, although we do not know to what extent the academic frustrations led students to alcohol and to what extent the alcohol use led to the academic troubles (Wood, Sher, Erickson, & DeBord, 1997).

Culture also plays an important role. For example, most Jewish families emphasize drinking in moderation, and relatively few Jews become alcoholics (Cahalan, 1978). The same conclusion holds true for Italians as well. By contrast, the Irish tend to be more tolerant of heavy drinking, and alcoholism is indeed more prevalent among people of Irish background (Vaillant & Milofsky, 1982).

Still, individuals differ. Not all children of alcoholic parents become alcoholics themselves, and not all children who grow up in a culture that tolerates heavy drinking become alcoholics. How can we predict which people are most vulnerable to alcoholism?

TABLE 16.3 Type I and Type II Alcoholism

	SEVERITY	GENDER DISTRIBUTION	GENETIC BASIS	ONSET
Type I (or Type A)	Generally less severe; better long-term outcome	Both males and females	Weaker genetic contribution	Gradual onset later in life
Type II (or Type B)	More severe, more likely to be associated with aggressive behavior and antisocial personality	Almost exclusively males	Strong evidence for genetic contribution	Rapid onset in teens or early 20s

WHAT'S THE EVIDENCE?

Ways of Predicting Alcoholism

Perhaps people's early behavior might offer some indicator of who is more or less likely to become an alcoholic. One way to find such clues would be to record the presence or absence of various behaviors in hundreds of young people; 20 years later, we could find out which of them have become alcoholics and determine which early behaviors might have predicted those outcomes. But such a study would take 20 years to complete. Moreover, it might be difficult to find some of the subjects after all that time, especially the alcoholics.

A more feasible approach would be to compare children of an alcoholic parent with children of parents who are not alcoholics. From previous studies, we know that more children of alcoholics will become alcoholics. Therefore, behaviors that are significantly more prevalent among the children of alcoholics may predict vulnerability to alcoholism.

In the first of the following studies, experimenters tested whether alcohol might be more rewarding to the sons of alcoholics than to the sons of nonalcoholics (Levenson, Oyama, & Meek, 1987). (The study focused on men because alcoholism is about twice as common in men as in women.)

EXPERIMENT 1

Hypothesis When people are placed in a stressful situation, the opportunity to drink alcohol will reduce stress for almost everyone. It will have a greater effect on the adult sons of an alcoholic parent than on other men of the same age.

Method The experiment was conducted on young men; half of them were sons of an alcoholic father, and half of them sons of nonalcoholic parents. The men were told that, at a certain time, they would receive an electric shock and, at another time, they would have to give a 3-minute speech entitled "what I like and dislike about my body." They

watched a clock tick off the waiting time. Half of each group were given alcohol to drink at the start of the waiting period, and everyone who was offered alcohol drank it.

Results All of the men showed considerable stress, as measured by heart rate, restlessness, and self-reports of emotions. All of those who drank alcohol showed a lower heart rate and reported less anxiety. The easing of stress was more pronounced in those who had an alcoholic father (Figure 16.11).

Interpretation Men who are genetically vulnerable to alcoholism experience greater stress-reducing effects from alcohol than other men of the same age. Perhaps the degree to which alcohol relieves stress may provide a measure of vulnerability to alcoholism.

Several other studies tested and confirmed the hypothesis that young men who are vulnerable to alcoholism might have difficulty estimating their own degree of intoxication. That is, those who are most vulnerable to alcoholism may not know when to stop. The following study tested whether young drinkers who underestimate their degree of intoxication are more likely than others to become alcoholics later in life (Schuckit & Smith, 1997).

EXPERIMENT 2

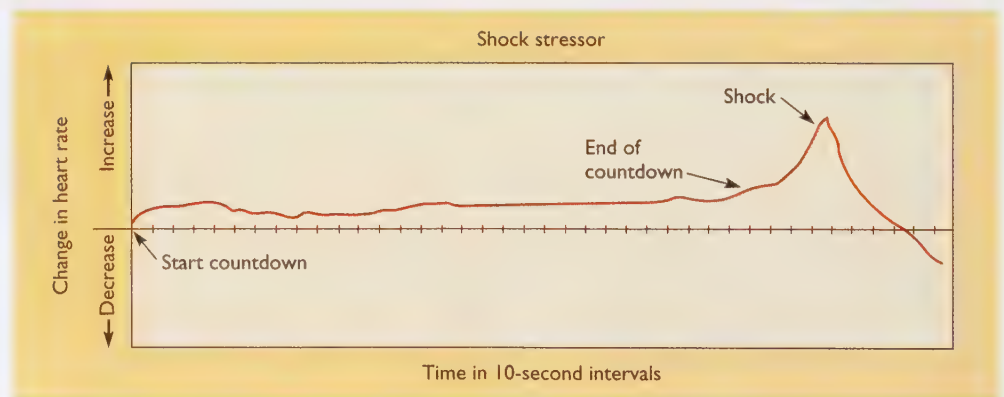
Hypothesis Men who underestimate how intoxicated they are after moderate drinking will be more likely than others to become alcoholics later.

Method This study was limited to 18- to 25-year-old men who had a close relative who was alcoholic. All of them drank a fixed amount of alcohol and were then asked to walk and to describe how intoxicated they felt. Experimenters noted how much the men staggered or swayed when they walked. Ten years later, the experimenters located as many of these men as possible, interviewed them, and determined whether they had become alcoholics.

Results Of those who either did not sway much when walking or stated that they did not feel intoxicated, 51 of 81 (63%) became alcoholics within 10 years. Of those who clearly swayed and reported that they felt intoxicated, only 9 of 52 (17%) became alcoholics.

FIGURE 16.11

Changes in stress over time for a typical subject: The line goes up to indicate an increase in heart rate. Note that heart rate increased as soon as the countdown began and then remained stable. It rose toward the end of the countdown and again at the time of the shock or speech. Alcohol suppressed these signs of stress, especially for the sons of alcoholics. (From Levenson, Oyama, & Meek, 1987.)



Interpretation Men who neither act nor feel intoxicated after a moderate amount of drinking are likely to continue drinking and to become alcoholics. By watching and interviewing young drinkers, psychologists may be able to identify a high-risk population.

Any research study has limitations, and one limitation of the above studies is clear: They examined only men. Considering that more men than women become alcoholics, the research strategy was reasonable; however, someone should repeat these studies with women.



Treatments for Alcoholism and Drug Addiction

Of all the people who try to quit alcohol or other drugs on their own, an estimated 10–20% manage to succeed (Cohen et al., 1989), though not necessarily on the first try. Some people quit and relapse and quit and relapse many times before eventually succeeding. In many cases, however, people with a substance abuse problem find that they cannot quit on their own. Eventually, they “hit bottom,” discovering that they have damaged their health, their ability to hold a job, and their relationships with friends and family. At that point, they turn to others for help—either a mental-health professional or a self-help group such as Alcoholics Anonymous.

People who seek professional help improve their likelihood of long-term abstinence, although there are no guarantees. Addicts who check into a hospital for treatment can be supervised 24 hours a day to ensure full abstinence. **Detoxification** refers to a supervised period to remove drugs from the body. In the long run, however, most addicts respond just as well to outpatient treatment as they do to hospital treatment (Miller & Hester, 1986).

Treating Alcoholism

*My mind is a dark place,
and I should not be left alone there at night.*

—PARTICIPANT AT ALCOHOLICS ANONYMOUS MEETING

Combating alcoholism is difficult but not hopeless. Let's consider several approaches and controversies.

Alcoholics Anonymous The most widespread treatment for alcoholism in North America is **Alcoholics Anonymous (AA)**, a self-help group of people who are trying to abstain from alcohol use and to help others do the same. AA meetings are held regularly in community halls, church basements, and other available spaces (Figure 16.12). The meeting format varies but often includes study of the book *Alcoholics Anonymous* (Anonymous, 1955) and discussions of participants' individual problems; some meetings feature an invited speaker. The group has a strong spiritual focus, including a reliance on “a Power greater



FIGURE 16.12 Alcoholics Anonymous (AA) is the prototypical example of a self-help group. AA meetings are held throughout the United States and in many other countries throughout the world. Members share their experiences with one another and provide help and encouragement to members in need. A member who is fighting a craving for alcohol can call a fellow member for help at any time; each understands the problems that the others are having. AA makes many recommendations, but its only requirement is that members make an effort to overcome their alcohol problems.

than ourselves,” although AA has no affiliation with any particular religion. Although AA imposes no requirements on its members other than making an effort to give up drinking alcohol, new members are strongly encouraged to attend 90 meetings during the first 90 days. (Those who miss a day can compensate by attending two or more meetings another day.) From then on, members attend as often as they like.

Millions of people have participated in the AA program in the United States and elsewhere around the world. One reason for its appeal is that all its members have gone through similar experiences. If someone tries to make an excuse for drinking, saying “you just don't understand how I feel,” others can retort, “oh, yes we do!” A member who feels the urge to take a drink or who has already had one, can call a fellow member day or night for support. There is no charge for attendance at meetings; members simply contribute what they can toward the cost of renting the meeting place. AA has inspired other “anonymous” self-help groups whose purpose is to help drug addicts, compulsive gamblers, compulsive eaters, and so forth.

Although AA members themselves have confidence in the value of the program, research on its effectiveness has been scarce. One reason is that the organization is serious about its members' anonymity; it does not provide a list of members, and many of its meetings are closed to nonmembers. One further problem is that people cannot be assigned randomly to an AA group and a control group; AA participants are simply those who wish to participate, and they undoubtedly differ in many ways from those who choose not to participate.

Antabuse In addition to or instead of attendance at AA meetings, some alcoholics seek medical treatment. Many years ago, investigators noticed that the workers in a certain

rubber manufacturing plant drank very little alcohol. The investigators eventually linked this behavior to *disulfiram*, a chemical that was used in the manufacturing process. Ordinarily, the liver converts alcohol into a toxic substance, *acetaldehyde* (ASS-eh-TAL-de-HIDE) and then converts acetaldehyde into a harmless substance, *acetic acid*. Disulfiram, however, blocks the conversion of acetaldehyde to acetic acid. Whenever the workers drank alcohol, however, acetaldehyde accumulated in their body, and they became ill. Over time, they learned to avoid all use of alcohol.

Disulfiram, available under the trade name **Antabuse**, is now commonly used in the treatment of alcoholism (Peachey & Naranjo, 1983). Alcoholics who take a daily Antabuse pill become very sick whenever they have a drink. They develop a sensation of heat in the face, headache, nausea, blurred vision, and anxiety. The threat of sickness is probably more effective than the sickness itself (Fuller & Roth, 1979). By taking a daily Antabuse pill, a recovering alcoholic renews the decision not to drink. Those who actually do take a drink in spite of the threat get quite ill, at which point they may either decide not to drink again or they may decide not to take the pill again!

CONCEPT CHECK

8. About 50% of Southeast Asians have a gene that makes them unable to convert acetaldehyde to acetic acid. Would such people be more likely or less likely than others to become alcoholics? (Check your answer on page 613.)

Is Substance Abuse a “Disease”? You have no doubt heard people say that alcoholism or drug abuse is a disease. It is hard to confirm or deny that statement, however, unless someone specifies exactly what the term *disease* means. (The medical profession gives the term no precise meaning, and the term has a wide variety of connotations.) When people call alcoholism or drug abuse a disease, they apparently mean that alcoholics and drug abusers should feel no guiltier about their condition than they would feel about having pneumonia.

Although the “disease” concept is more constructive than thinking of substance abuse as a sign of moral weakness, the concept has some implications that the data do not support. For example, it implies an all-or-none distinction between those who have the disease and those who do not. The current trend among psychologists favors a continuum from people with no problem to those with a severe problem, including every possible intermediate step (Miller & Brown, 1997; Polcin, 1997).

Also, the “disease” concept implies that alcoholism or drug abuse becomes inevitably worse over time. In fact, the long-term outcome for alcoholics and drug abusers varies enormously (Hser, Anglin, & Powers, 1993; Vaillant, 1983). As Figure 16.13 shows, some individuals deteriorate rapidly and severely, some reach a steady level of abuse, and still others show a gradual improvement in their condition over time.

Furthermore, to regard substance abuse strictly as a disease seems to imply a need for a medical intervention and downplays the importance of environmental factors.

FIGURE 16.13 Researchers studied 581 young adult male narcotics addicts in 1962–1964. They asked these men about their previous drug use and located most of them again in later years to check on their progress. Over time, the results varied. Some died or went to prison; some continued using drugs as before; some decreased their use of drugs; and some abstained altogether. (From Hser, Anglin, & Powers, 1993.)

